In [22]:
```
import pandas as pd
import seaborn as sns
import numpy as np
from matplotlib import pyplot as plt
```

In [7]:
```
fifa =pd.read_csv('fifa.csv')
```

In [8]:
```
fifa.head()
```

Out[8]:

	sofifa_id	player_url	short_name	long_name	age	dob	height_cm	weight_kg	nationality	club	...
0	158023	https://sofifa.com/player/158023/lionel-messi/...	L. Messi	Lionel Andrés Messi Cuccittini	32	1987-06-24	170	72	Argentina	FC Barcelona	...
1	20801	https://sofifa.com/player/20801/c-ronaldo-dos-...	Cristiano Ronaldo	Cristiano Ronaldo dos Santos Aveiro	34	1985-02-05	187	83	Portugal	Juventus	...
2	190871	https://sofifa.com/player/190871/neymar-da-sil...	Neymar Jr	Neymar da Silva Santos Junior	27	1992-02-05	175	68	Brazil	Paris Saint-Germain	...
3	200389	https://sofifa.com/player/200389/jan-oblak/20/...	J. Oblak	Jan Oblak	26	1993-01-07	188	87	Slovenia	Atlético Madrid	...
4	183277	https://sofifa.com/player/183277/eden-hazard/2...	E. Hazard	Eden Hazard	28	1991-01-07	175	74	Belgium	Real Madrid	...

5 rows × 104 columns

In [9]:
```
for col in fifa.columns:
    print(col)
```

```
sofifa_id
player_url
short_name
long_name
age
dob
height_cm
weight_kg
nationality
club
overall
potential
value_eur
wage_eur
player_positions
preferred_foot
international_reputation
weak_foot
skill_moves
work_rate
body_type
real_face
release_clause_eur
player_tags
team_position
team_jersey_number
loaned_from
joined
contract_valid_until
nation_position
nation_jersey_number
pace
shooting
passing
dribbling
defending
physic
gk_diving
gk_handling
gk_kicking
gk_reflexes
```

```
gk_speed
gk_positioning
player_traits
attacking_crossing
attacking_finishing
attacking_heading_accuracy
attacking_short_passing
attacking_volleys
skill_dribbling
skill_curve
skill_fk_accuracy
skill_long_passing
skill_ball_control
movement_acceleration
movement_sprint_speed
movement_agility
movement_reactions
movement_balance
power_shot_power
power_jumping
power_stamina
power_strength
power_long_shots
mentality_aggression
mentality_interceptions
mentality_positioning
mentality_vision
mentality_penalties
mentality_composure
defending_marking
defending_standing_tackle
defending_sliding_tackle
goalkeeping_diving
goalkeeping_handling
goalkeeping_kicking
goalkeeping_positioning
goalkeeping_reflexes
ls
st
rs
lw
```

```
lf
cf
rf
rw
lam
cam
ram
lm
lcm
cm
rcm
rm
lwb
ldm
cdm
rdm
rwb
lb
lcb
cb
rcb
rb
```

In [10]:
```
fifa.shape
```

Out[10]:
```
(18278, 104)
```

In [11]:
```
18278*104
```

Out[11]:
```
1900912
```

In [13]:
```
fifa['nationality'].value_counts()[0:10]
```

Out[13]:
```
nationality
England        1667
Germany        1216
Spain          1035
France          984
Argentina       886
Brazil          824
Italy           732
Colombia        591
Japan           453
Netherlands     416
Name: count, dtype: int64
```

In [15]:
```
fifa['nationality'].value_counts()[0:5].keys()
```

Out[15]:
```
Index(['England', 'Germany', 'Spain', 'France', 'Argentina'], dtype='object', name='nationality')
```

In [29]:
```
# Data Cleaning
# Checking for missing values
fifa.isnull().sum()

# Drop irrelevant columns
fifa.drop(columns=['sofifa_id', 'player_url'], inplace=True, errors='ignore')

# Handle missing values
fifa.fillna({'club': 'Unknown'}, inplace=True)
```

In [32]:
```
# overview
fifa.shape
fifa['nationality'].nunique()
fifa['club'].nunique()
```

Out[32]:
```
698
```

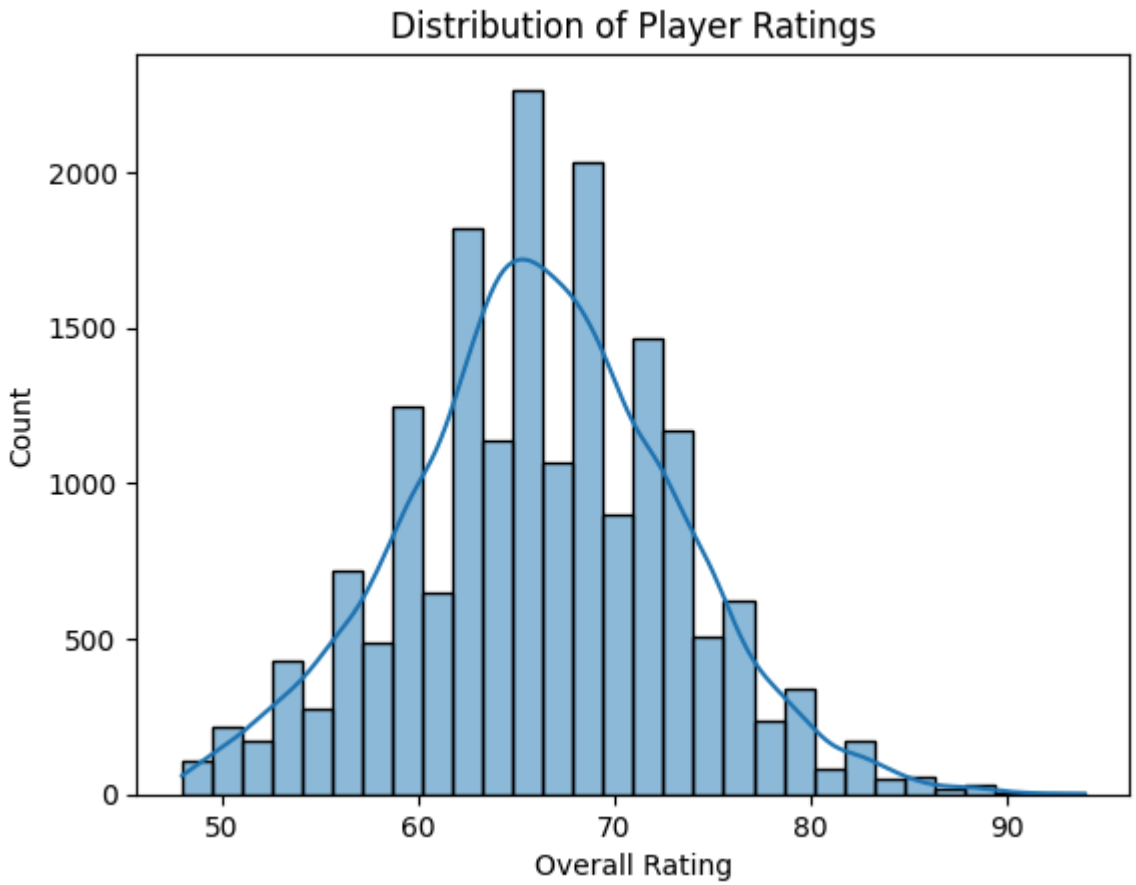
In [33]:
```
# top 10 players
top_players = fifa[['short_name', 'overall', 'club']].sort_values(by='overall', ascending=False).head(10)
top_players
```

Out[33]:

	short_name	overall	club
0	L. Messi	94	FC Barcelona
1	Cristiano Ronaldo	93	Juventus
2	Neymar Jr	92	Paris Saint-Germain
3	J. Oblak	91	Atlético Madrid
4	E. Hazard	91	Real Madrid
5	K. De Bruyne	91	Manchester City
6	M. ter Stegen	90	FC Barcelona
8	L. Modrić	90	Real Madrid
7	V. van Dijk	90	Liverpool
9	M. Salah	90	Liverpool

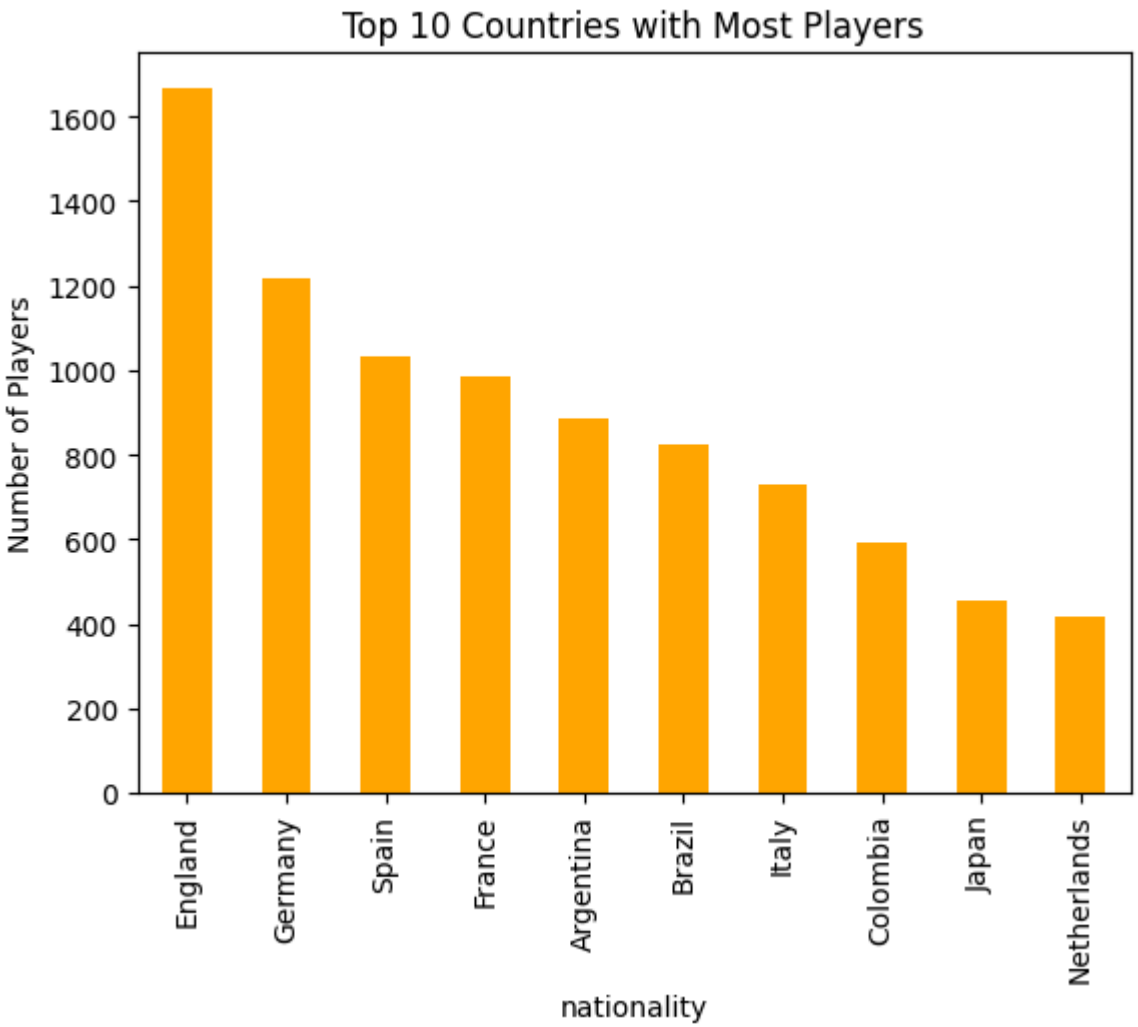
In [35]:

```
# players rating
sns.histplot(fifa['overall'], bins=30, kde=True)
plt.title("Distribution of Player Ratings")
plt.xlabel("Overall Rating")
plt.ylabel("Count")
plt.show()
```

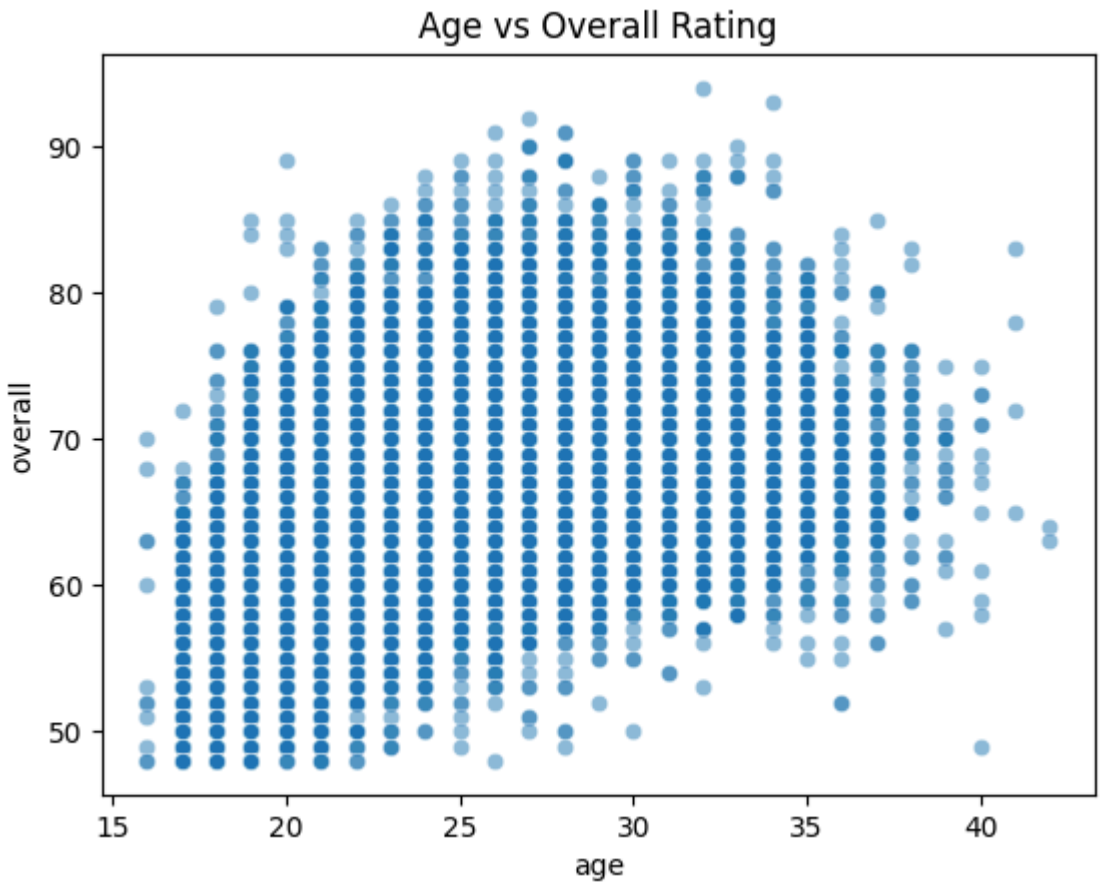


In [ ]:

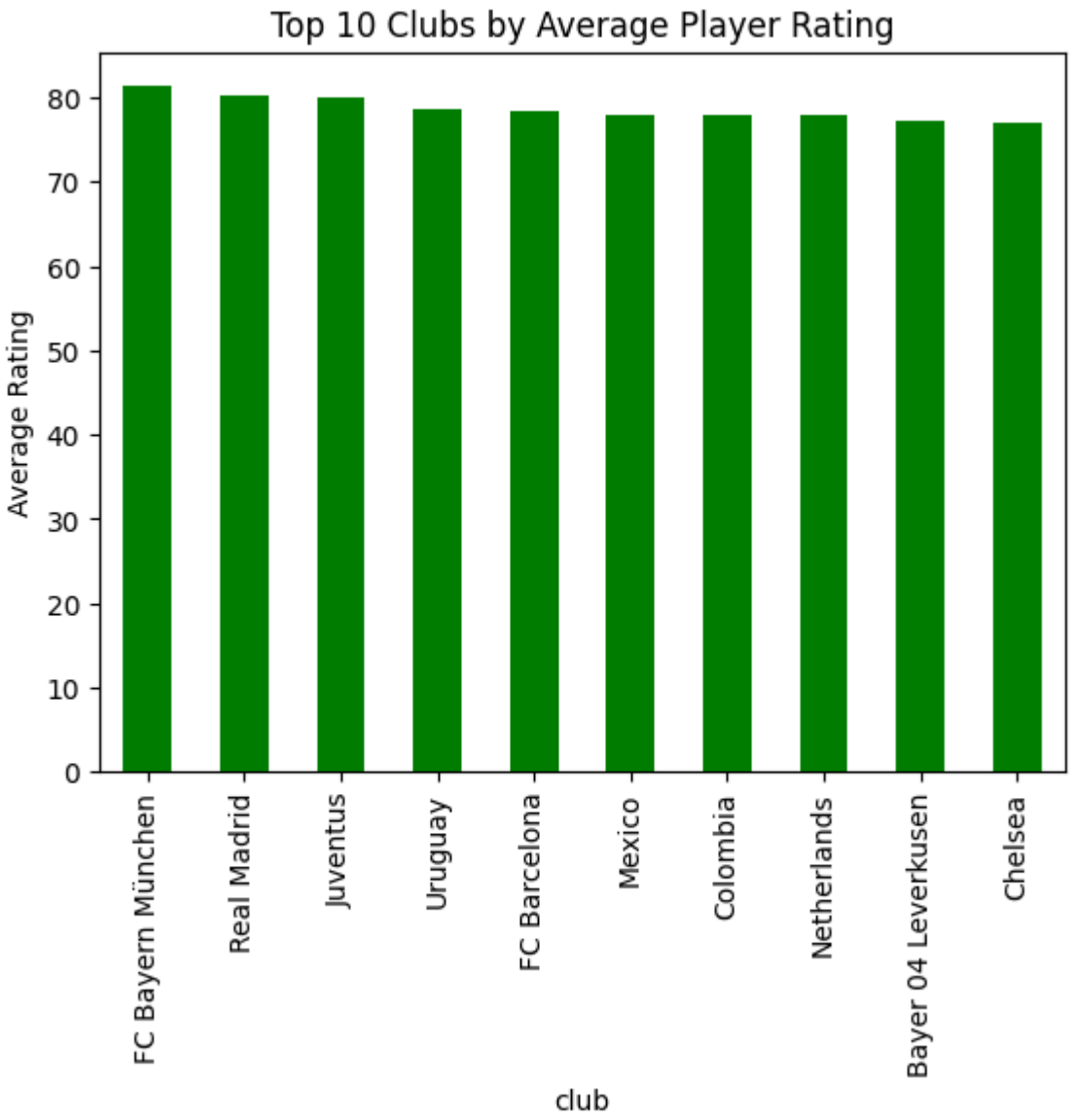
In [36]:
```
# top 10 countries with most players
fifa['nationality'].value_counts().head(10).plot(kind='bar', color='orange')
plt.title("Top 10 Countries with Most Players")
plt.ylabel("Number of Players")
plt.show()
```



In [39]:
```
sns.scatterplot(x='age', y='overall', data=fifa, alpha=0.5)
plt.title("Age vs Overall Rating")
plt.show()
```



In [40]:
```
club_ratings = fifa.groupby('club')['overall'].mean().sort_values(ascending=False).head(10)
club_ratings.plot(kind='bar', color='green')
plt.title("Top 10 Clubs by Average Player Rating")
plt.ylabel("Average Rating")
plt.show()
```



In [43]:
```
# Young Players
young_talents = fifa[fifa['overall'] > 85].sort_values('age').head(5)
young_talents[['short_name', 'age', 'overall']]
```

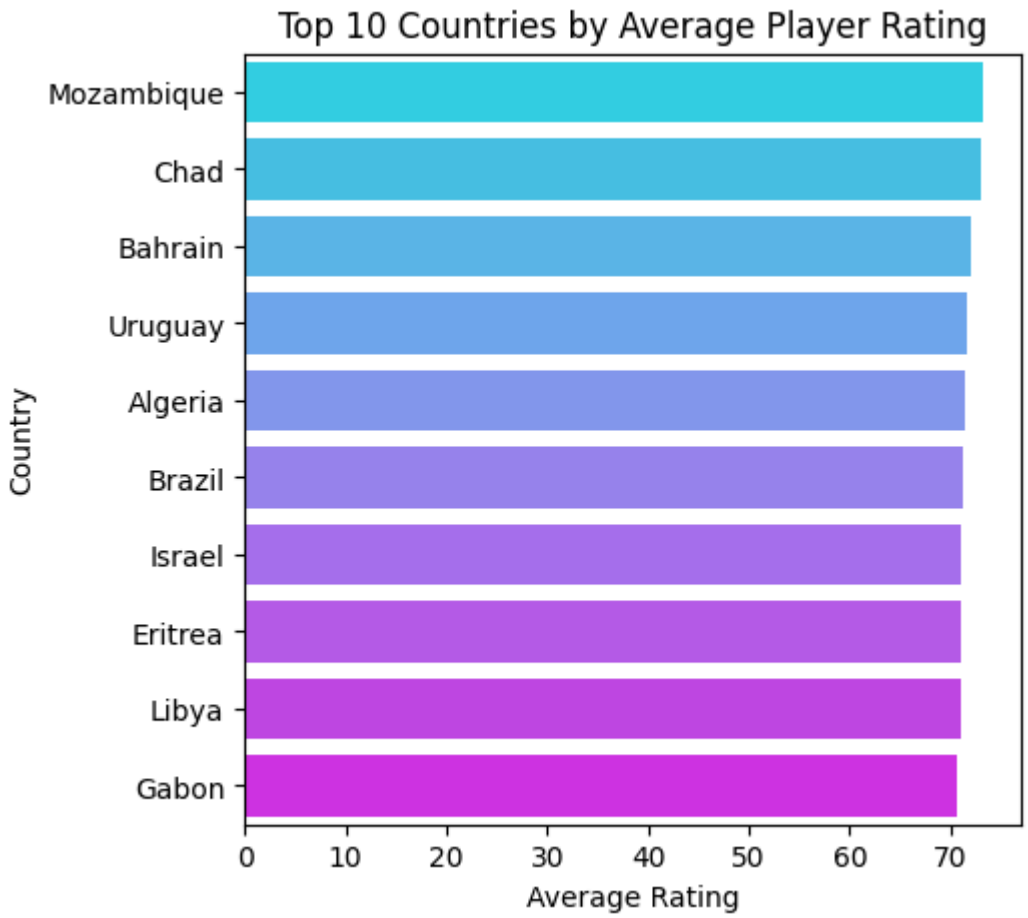
Out[43]:

	short_name	age	overall
10	K. Mbappé	20	89
55	L. Sané	23	86
41	Bernardo Silva	24	87
26	R. Sterling	24	88
61	J. Kimmich	24	86

In [48]:

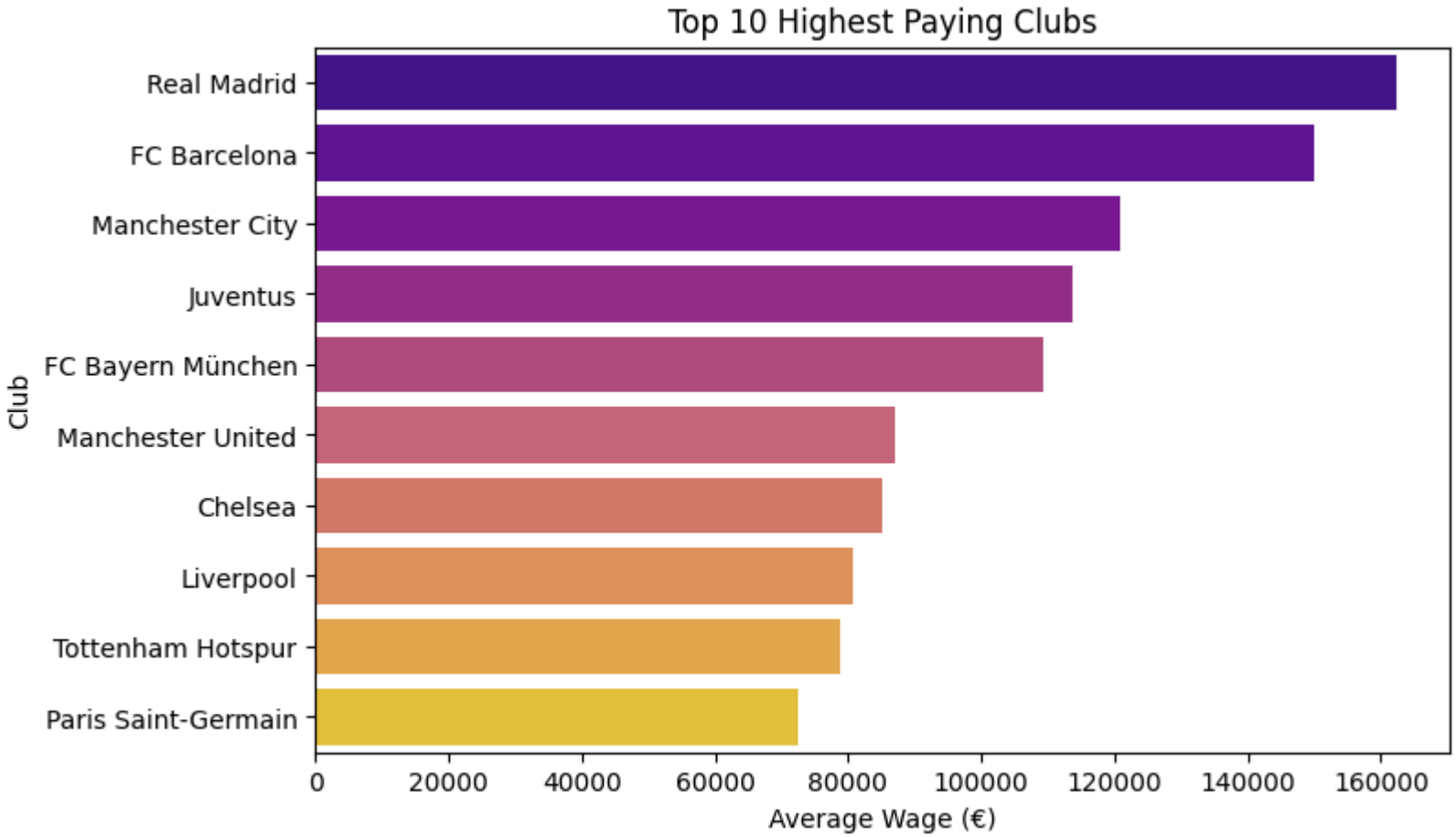
```
avg_rating_country = fifa.groupby('nationality')['overall'].mean().sort_values(ascending=False).head(10)
plt.figure(figsize=(5,5))
sns.barplot(x=avg_rating_country.values, y=avg_rating_country.index, palette='cool')
plt.title("Top 10 Countries by Average Player Rating")
plt.xlabel("Average Rating")
plt.ylabel("Country")
plt.show()
```

```
C:\Users\Hanif\AppData\Local\Temp\ipykernel_7740\1684700414.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and se
t `legend=False` for the same effect.

  sns.barplot(x=avg_rating_country.values, y=avg_rating_country.index, palette='cool')
```



In [53]:
```
highest_paying_clubs = fifa.groupby('club')['wage_eur'].mean().sort_values(ascending=False).head(10)
plt.figure(figsize=(8,5))
sns.barplot(x=highest_paying_clubs.values, y=highest_paying_clubs.index, palette='plasma')
plt.title("Top 10 Highest Paying Clubs")
plt.xlabel("Average Wage (€)")
plt.ylabel("Club")
plt.show()
```

```
C:\Users\Hanif\AppData\Local\Temp\ipykernel_7740\4285414960.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and se
t `legend=False` for the same effect.

  sns.barplot(x=highest_paying_clubs.values, y=highest_paying_clubs.index, palette='plasma')
```



In [54]:
```
germany = fifa[fifa['nationality'] == 'Germany']

# Tallest German Players
germany.sort_values('height_cm', ascending=False).head(5)
```

```
# Heaviest German Players
germany.sort_values('weight_kg', ascending=False).head(5)

# Top German Players by Wage
germany.sort_values('wage_eur', ascending=False).head(5)
```

Out[54]:

	short_name	long_name	age	dob	height_cm	weight_kg	nationality	club	overall	potential	...	lwb	ldm	cdm	rdm
36	T. Kroos	Toni Kroos	29	1990-01-04	183	76	Germany	Real Madrid	88	88	...	75+3	79+3	79+3	79+3
6	M. ter Stegen	Marc-André ter Stegen	27	1992-04-30	187	85	Germany	FC Barcelona	90	93	...	NaN	NaN	NaN	NaN
55	L. Sané	Leroy Sané	23	1996-01-11	183	75	Germany	Manchester City	86	92	...	65+2	59+2	59+2	59+2
146	I. Gündoğan	İlkay Gündoğan	28	1990-10-24	180	80	Germany	Manchester City	84	84	...	76+3	79+3	79+3	79+3
70	T. Müller	Thomas Müller	29	1989-09-13	186	75	Germany	FC Bayern München	86	86	...	69+3	68+3	68+3	68+3

5 rows × 102 columns

In [55]:

```
real_madrid = fifa[fifa['club'] == 'Real Madrid']

# Top 5 shooters
real_madrid[['short_name', 'shooting']].sort_values('shooting', ascending=False).head(5)

# Top 5 defenders
real_madrid[['short_name', 'defending']].sort_values('defending', ascending=False).head(5)

# Player nationalities in Real Madrid
real_madrid['nationality'].value_counts()
```

Out[55]:
```
nationality
Spain                 13
Brazil                 6
France                 3
Belgium                2
Croatia                1
Germany                1
Costa Rica             1
Wales                  1
Colombia               1
Serbia                 1
Dominican Republic     1
Uruguay                1
Japan                  1
Name: count, dtype: int64
```

In [61]:
```
# Top 5 youngest players with overall rating > 85
youngest_high_rated = fifa[fifa['overall'] > 85].sort_values('age').head(5)

print(" Top 5 Youngest High-Rated Players:")
display(youngest_high_rated[['short_name', 'age', 'overall', 'club', 'nationality']])

# Calculate average wage per club
highest_paying_club = fifa.groupby('club')['wage_eur'].mean().sort_values(ascending=False).head(1)

print("Club Paying Highest Average Wages:")
display(highest_paying_club)
 # Average rating by nationality
highest_avg_rating_nation = fifa.groupby('nationality')['overall'].mean().sort_values(ascending=False).head(1)

print("Nationality with Highest Average Overall Rating:")
display(highest_avg_rating_nation)
```

```
Top 5 Youngest High-Rated Players:
```

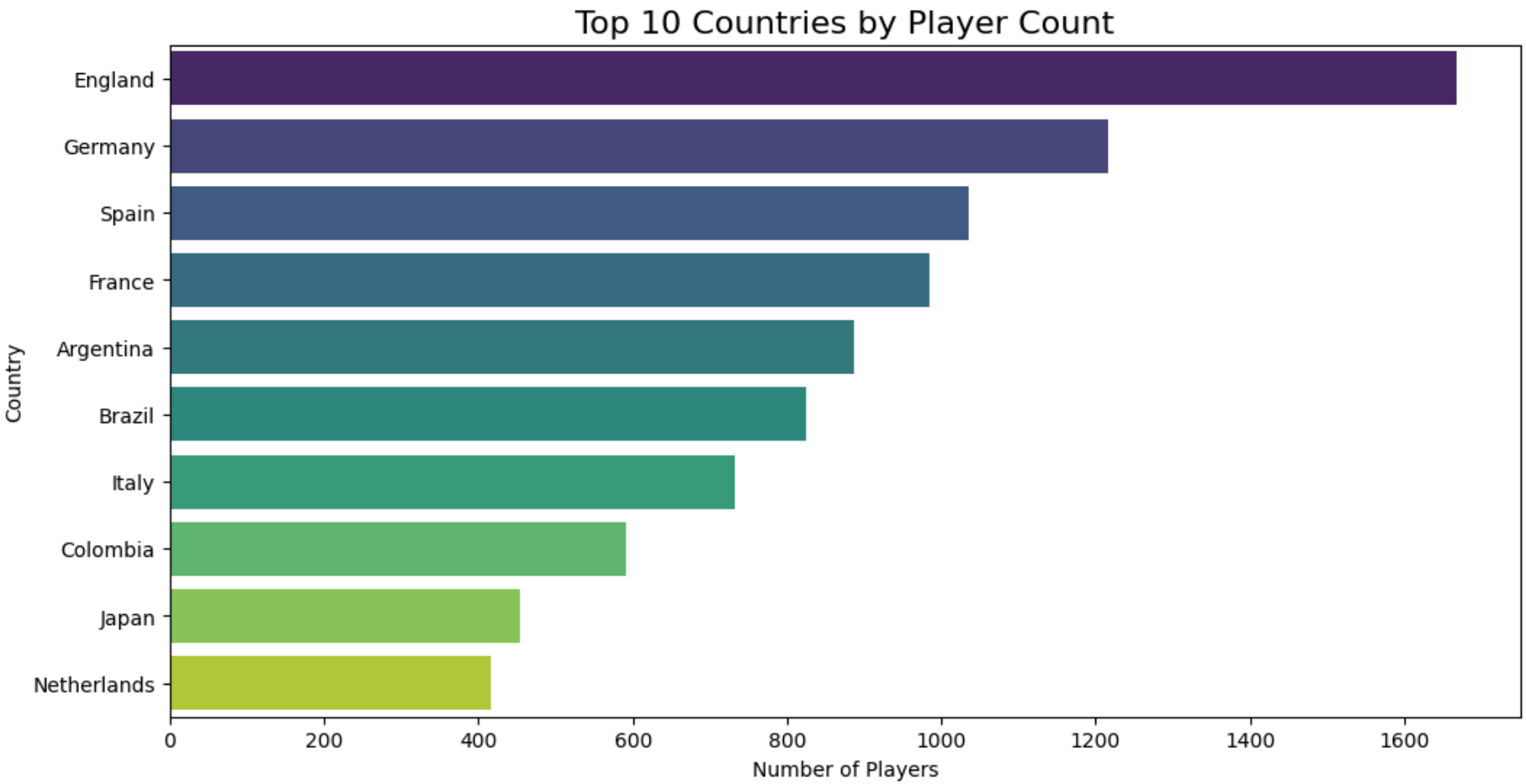
	short_name	age	overall	club	nationality
10	K. Mbappé	20	89	Paris Saint-Germain	France
55	L. Sané	23	86	Manchester City	Germany
41	Bernardo Silva	24	87	Manchester City	Portugal
26	R. Sterling	24	88	Manchester City	England
61	J. Kimmich	24	86	FC Bayern München	Germany

```
Club Paying Highest Average Wages:
club
Real Madrid    162242.424242
Name: wage_eur, dtype: float64
Nationality with Highest Average Overall Rating:
nationality
Mozambique    73.25
Name: overall, dtype: float64
```

In [62]:

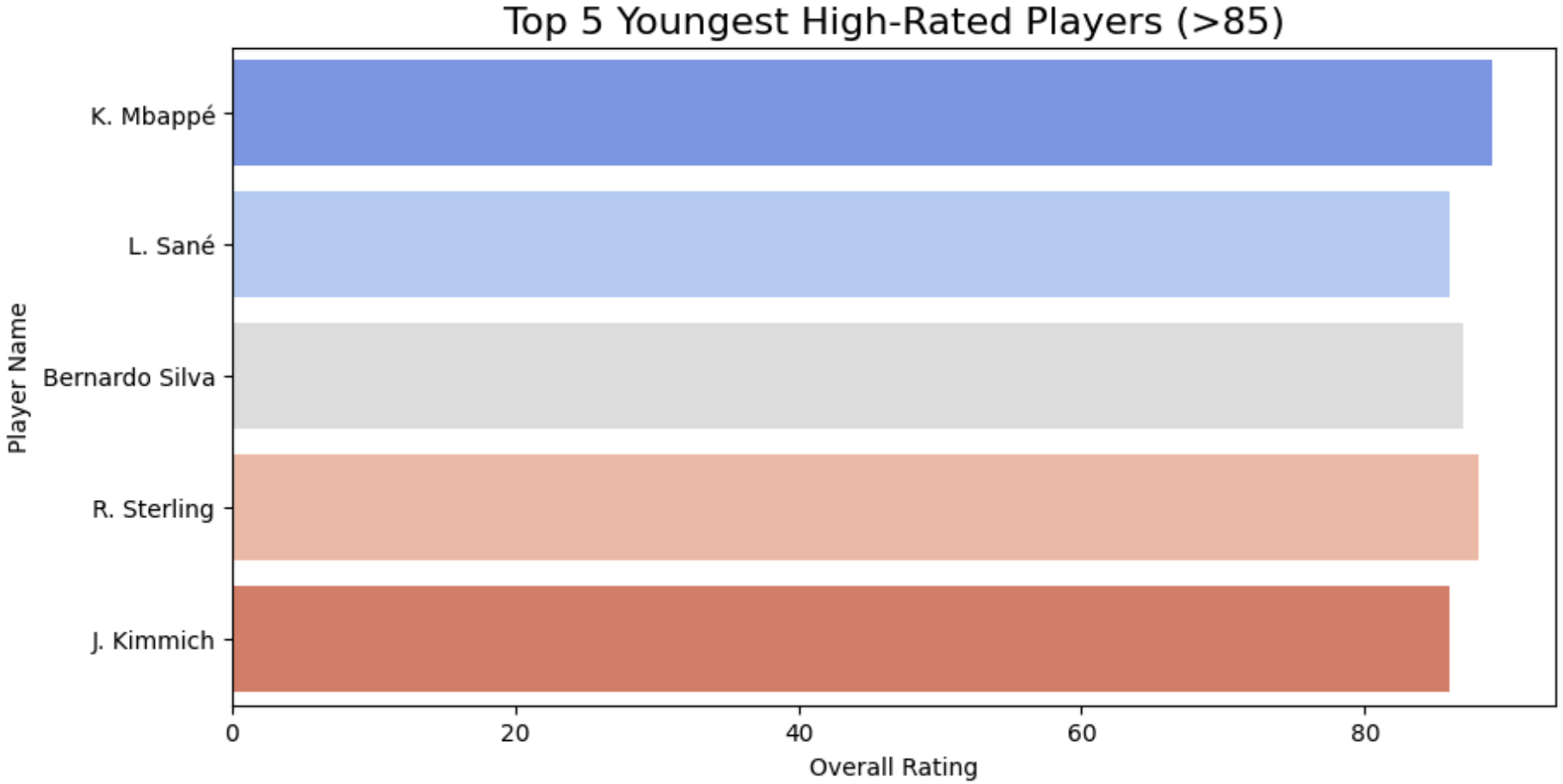
```
plt.figure(figsize=(12,6))
top_countries = fifa['nationality'].value_counts().head(10)
sns.barplot(x=top_countries.values, y=top_countries.index, palette='viridis')
plt.title("Top 10 Countries by Player Count", fontsize=16)
plt.xlabel("Number of Players")
plt.ylabel("Country")
plt.show()
```

```
C:\Users\Hanif\AppData\Local\Temp\ipykernel_7740\3760278816.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and se
t `legend=False` for the same effect.

  sns.barplot(x=top_countries.values, y=top_countries.index, palette='viridis')
```



In [63]:
```
youngest_high_rated = fifa[fifa['overall'] > 85].sort_values('age').head(5)

plt.figure(figsize=(10,5))
sns.barplot(x='overall', y='short_name', data=youngest_high_rated, palette='coolwarm')
plt.title("Top 5 Youngest High-Rated Players (>85)", fontsize=16)
plt.xlabel("Overall Rating")
plt.ylabel("Player Name")
plt.show()
```

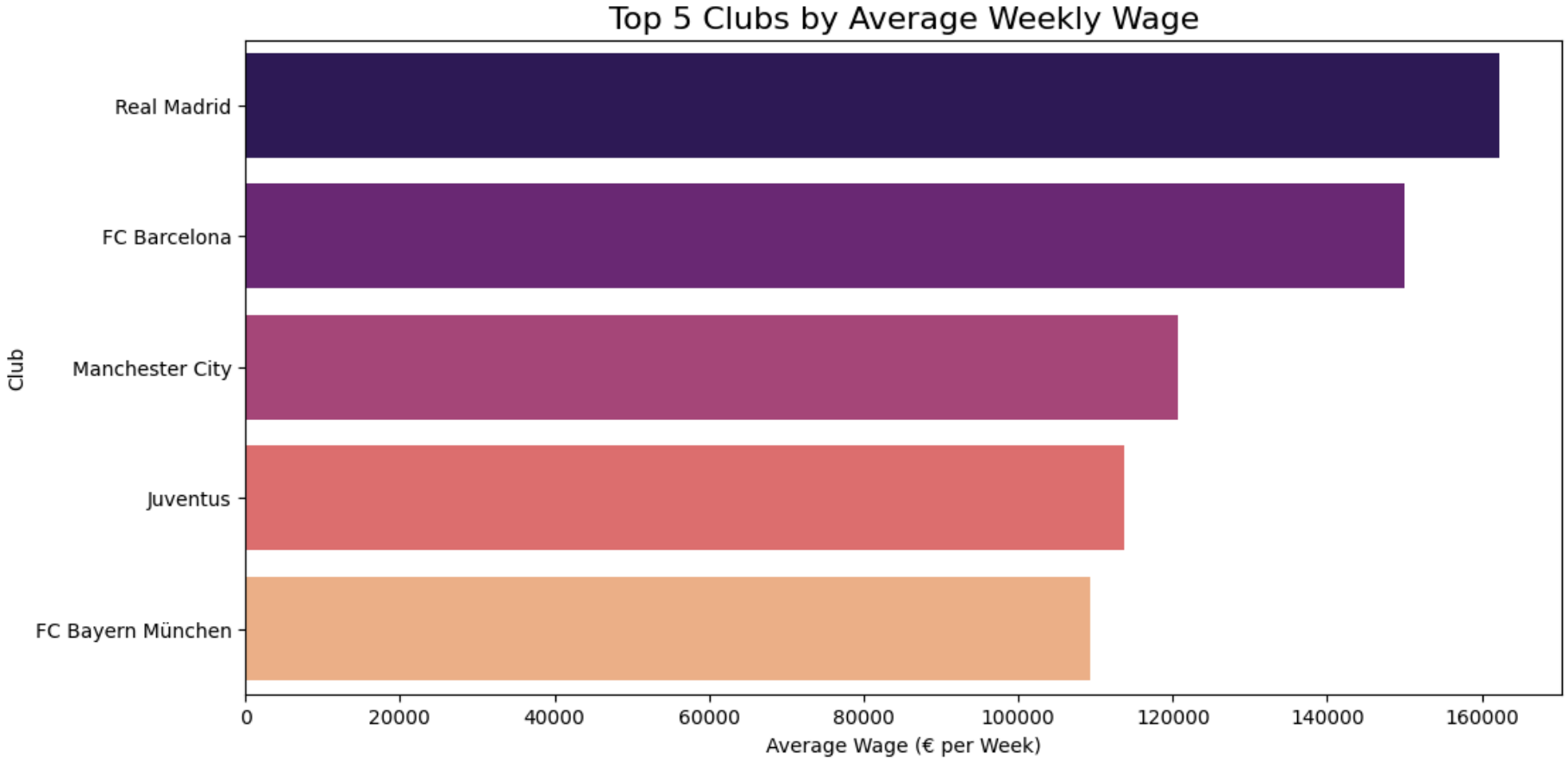
```
C:\Users\Hanif\AppData\Local\Temp\ipykernel_7740\2162645430.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and se
t `legend=False` for the same effect.

  sns.barplot(x='overall', y='short_name', data=youngest_high_rated, palette='coolwarm')
```



In [64]:
```
top_5_wage_clubs = fifa.groupby('club')['wage_eur'].mean().sort_values(ascending=False).head(5).reset_index()

plt.figure(figsize=(12,6))
sns.barplot(x='wage_eur', y='club', data=top_5_wage_clubs, palette='magma')
plt.title("Top 5 Clubs by Average Weekly Wage", fontsize=16)
plt.xlabel("Average Wage (€ per Week)")
```

```
plt.ylabel("Club")
plt.show()
```

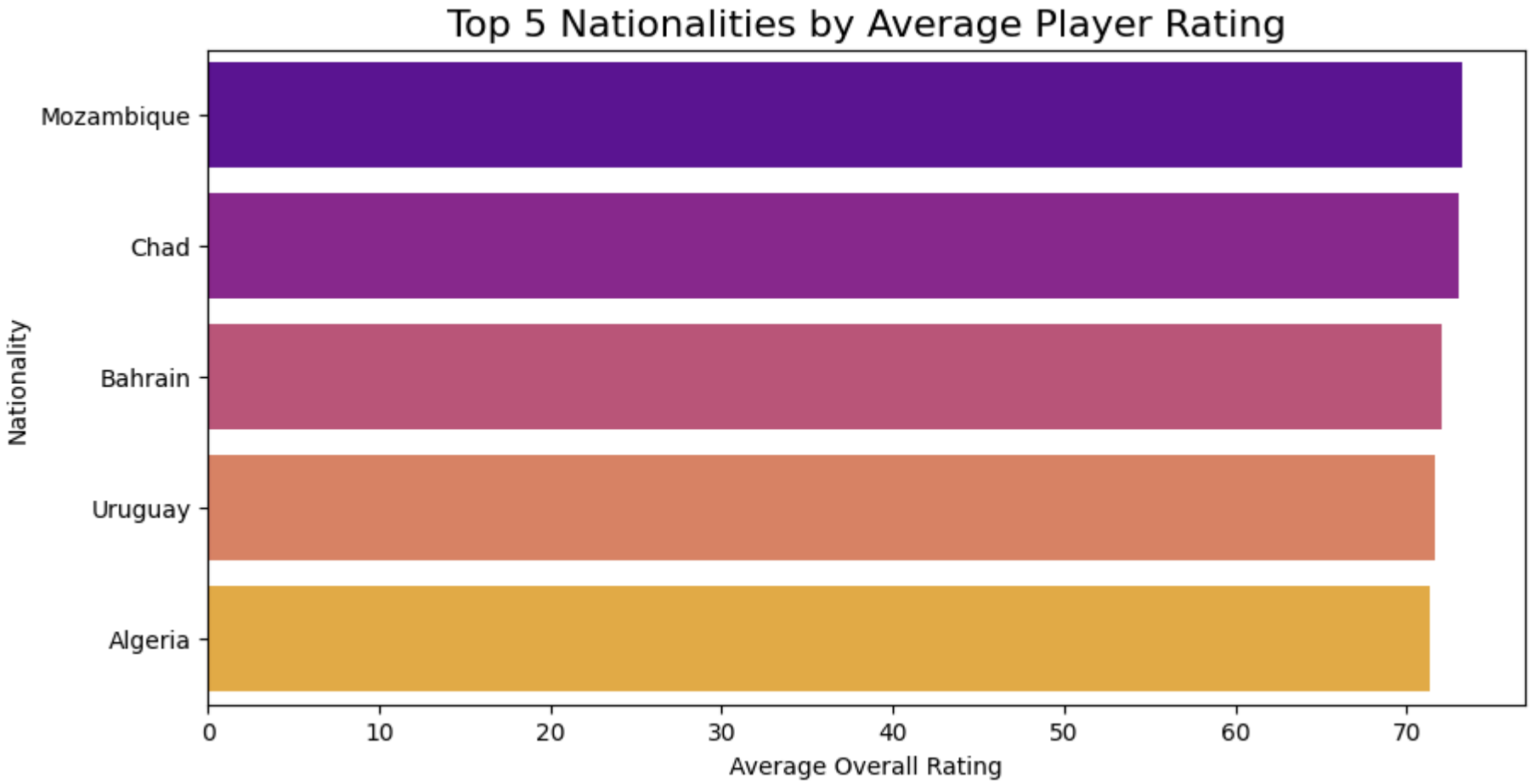
```
C:\Users\Hanif\AppData\Local\Temp\ipykernel_7740\3242603366.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and se
t `legend=False` for the same effect.

  sns.barplot(x='wage_eur', y='club', data=top_5_wage_clubs, palette='magma')
```



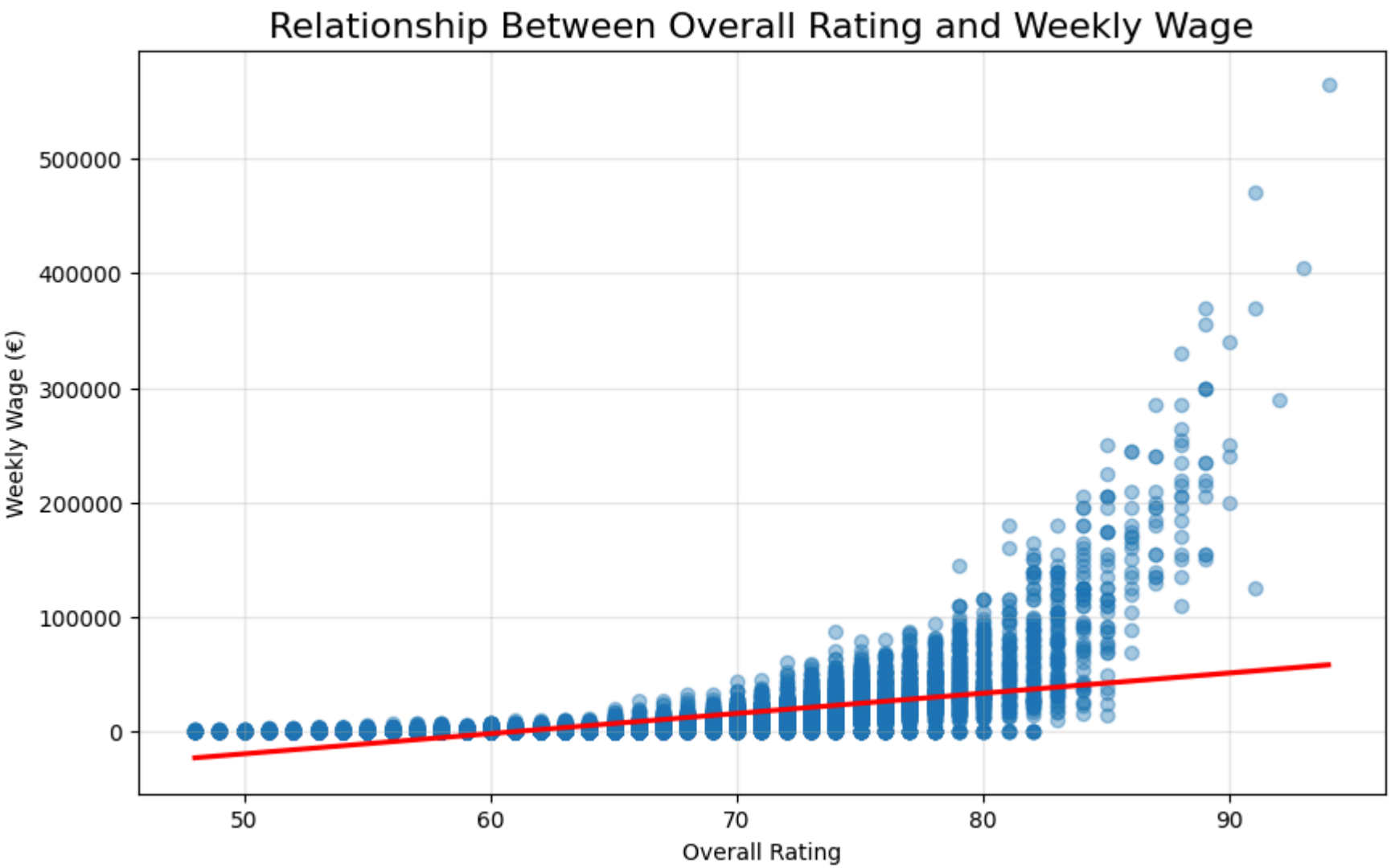
In [65]:
```
top_5_nations = fifa.groupby('nationality')['overall'].mean().sort_values(ascending=False).head(5).reset_index()

plt.figure(figsize=(10,5))
sns.barplot(x='overall', y='nationality', data=top_5_nations, palette='plasma')
```

```
plt.title("Top 5 Nationalities by Average Player Rating", fontsize=16)
plt.xlabel("Average Overall Rating")
plt.ylabel("Nationality")
plt.show()
```

```
C:\Users\Hanif\AppData\Local\Temp\ipykernel_7740\320333173.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and se
t `legend=False` for the same effect.

  sns.barplot(x='overall', y='nationality', data=top_5_nations, palette='plasma')
```



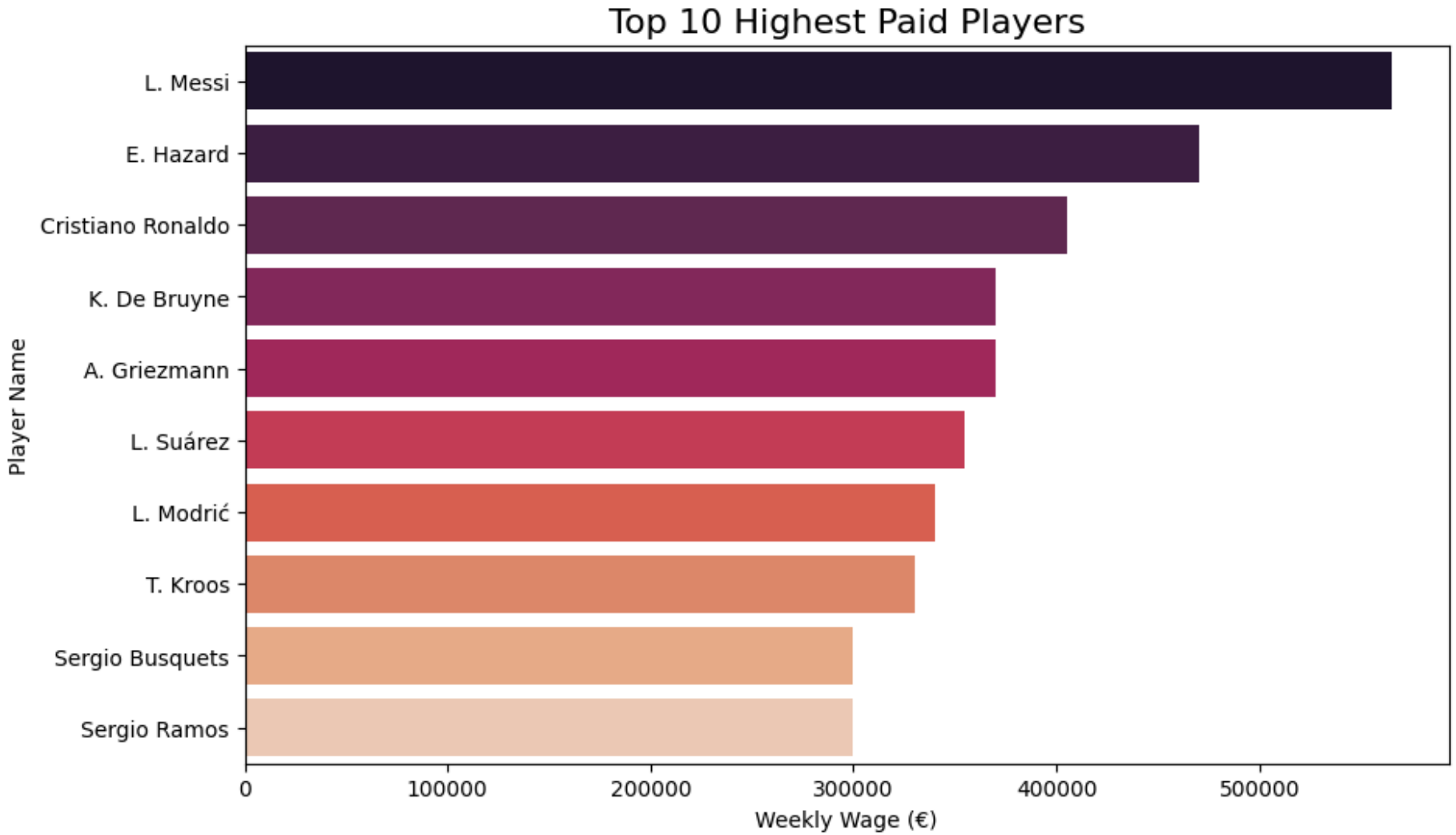
In [66]:
```
plt.figure(figsize=(10,6))
sns.regplot(x='overall', y='wage_eur', data=fifa, scatter_kws={'alpha':0.4}, line_kws={'color':'red'})
plt.title("Relationship Between Overall Rating and Weekly Wage", fontsize=16)
plt.xlabel("Overall Rating")
plt.ylabel("Weekly Wage (€)")
plt.grid(alpha=0.3)
plt.show()
```



In [67]:
```
top_paid = fifa[['short_name', 'wage_eur']].sort_values('wage_eur', ascending=False).head(10)

plt.figure(figsize=(10,6))
sns.barplot(x='wage_eur', y='short_name', data=top_paid, palette='rocket')
plt.title("Top 10 Highest Paid Players", fontsize=16)
plt.xlabel("Weekly Wage (€)")
```

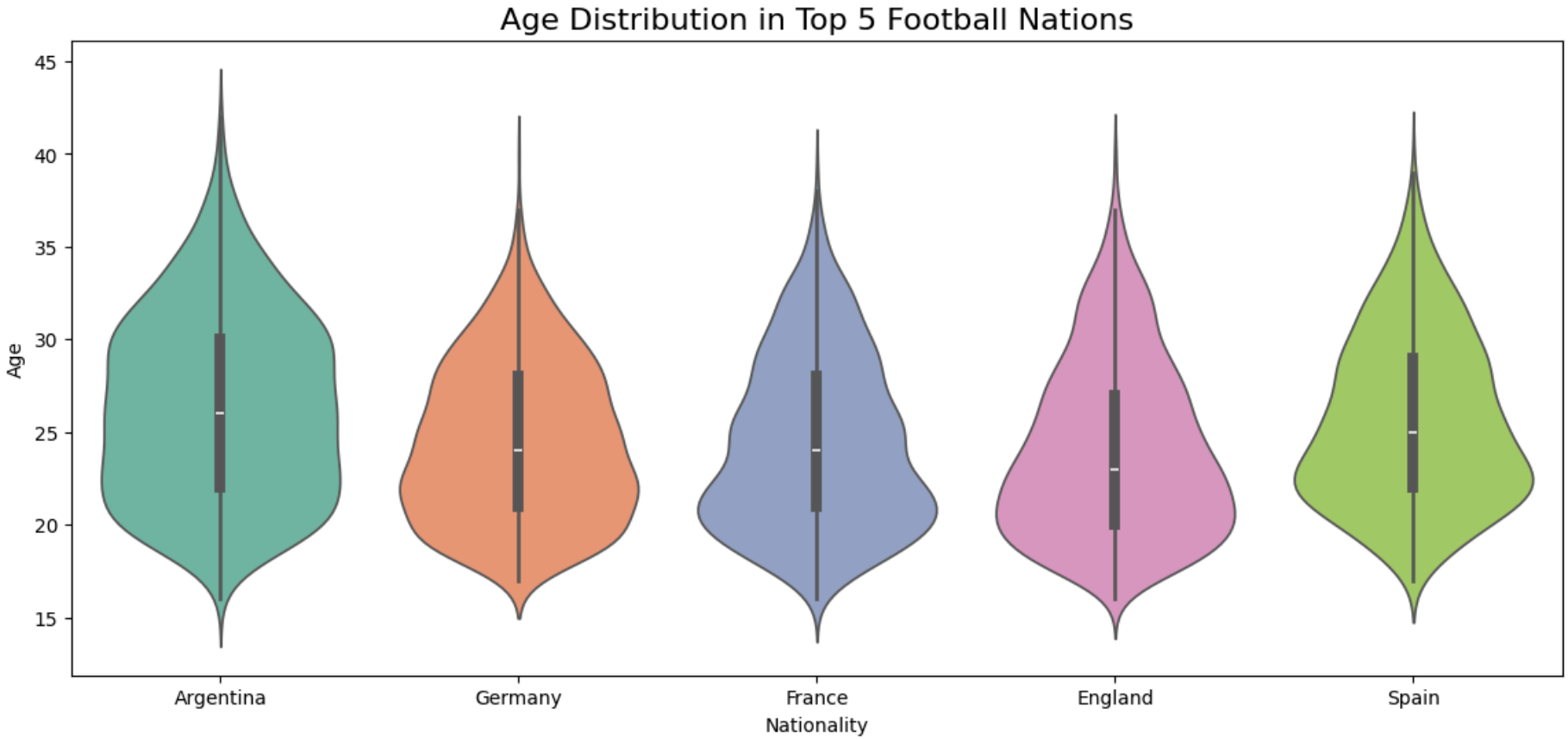
```
plt.ylabel("Player Name")
plt.show()
```

```
C:\Users\Hanif\AppData\Local\Temp\ipykernel_7740\422177916.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and se
t `legend=False` for the same effect.

  sns.barplot(x='wage_eur', y='short_name', data=top_paid, palette='rocket')
```



In [68]:
```
plt.figure(figsize=(14,6))
top_nations = fifa['nationality'].value_counts().head(5).index
sns.violinplot(x='nationality', y='age', data=fifa[fifa['nationality'].isin(top_nations)], palette='Set2')
plt.title("Age Distribution in Top 5 Football Nations", fontsize=16)
plt.xlabel("Nationality")
plt.ylabel("Age")
plt.show()
```

```
C:\Users\Hanif\AppData\Local\Temp\ipykernel_7740\4234239794.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and se
t `legend=False` for the same effect.

  sns.violinplot(x='nationality', y='age', data=fifa[fifa['nationality'].isin(top_nations)], palette='Set2')
```



In [70]:
```
# Convert 'overall' column to NumPy array
overall_array = fifa['overall'].to_numpy()

print("Shape:", overall_array.shape)
print("First 10 Ratings:", overall_array[:10])
```

```
Shape: (18278,)
First 10 Ratings: [94 93 92 91 91 91 90 90 90 90]
```

In [71]:
```
# Mean, Median, Std Dev of overall ratings
mean_rating = np.mean(overall_array)
median_rating = np.median(overall_array)
std_rating = np.std(overall_array)

print("Mean Overall Rating:", mean_rating)
print("Median Overall Rating:", median_rating)
print("Standard Deviation of Ratings:", std_rating)
```

```
Mean Overall Rating: 66.24499398183609
Median Overall Rating: 66.0
Standard Deviation of Ratings: 6.949763134372512
```

In [72]:
```
# Players with rating > 85
high_rated_mask = overall_array > 85
high_rated_players = fifa[high_rated_mask]

print("Number of High Rated Players (>85):", len(high_rated_players))
high_rated_players[['short_name', 'overall']].head()
```

```
Number of High Rated Players (>85): 73
```

Out[72]:

	short_name	overall
0	L. Messi	94
1	Cristiano Ronaldo	93
2	Neymar Jr	92
3	J. Oblak	91
4	E. Hazard	91

In [73]:

```
# Z-score for player ratings
z_scores = (overall_array - np.mean(overall_array)) / np.std(overall_array)

# Add Z-score back to DataFrame
fifa['rating_zscore'] = z_scores

fifa[['short_name', 'overall', 'rating_zscore']].head()
```

Out[73]:

	short_name	overall	rating_zscore
0	L. Messi	94	3.993662
1	Cristiano Ronaldo	93	3.849772
2	Neymar Jr	92	3.705883
3	J. Oblak	91	3.561993
4	E. Hazard	91	3.561993

In [74]:

```
# Randomly select 10 player ratings
random_ratings = np.random.choice(overall_array, size=10, replace=False)
print("Random Sample of Ratings:", random_ratings)
```

```
Random Sample of Ratings: [66 57 60 62 63 63 68 69 52 73]
```

In [75]:

```
# Increase all ratings by 2
boosted_ratings = overall_array + 2
```

```
# Compare original vs boosted
print("Original Ratings:", overall_array[:5])
print("Boosted Ratings :", boosted_ratings[:5])
```

```
Original Ratings: [94 93 92 91 91]
Boosted Ratings : [96 95 94 93 93]
```

In [76]:

```
# Create a 2D array of random ratings
ratings_matrix = np.random.randint(60, 95, size=(5, 3))
print("Ratings Matrix:\n", ratings_matrix)

# Calculate mean rating per player
mean_per_player = np.mean(ratings_matrix, axis=1)
print("Mean per Player:", mean_per_player)

# Transpose the matrix
print("Transposed Matrix:\n", ratings_matrix.T)
```

```
Ratings Matrix:
 [[78 75 84]
 [74 67 63]
 [80 94 68]
 [94 81 83]
 [79 62 72]]
Mean per Player: [79.         68.         80.66666667 86.         71.        ]
Transposed Matrix:
 [[78 74 80 94 79]
 [75 67 94 81 62]
 [84 63 68 83 72]]
```

In [77]:

```
# Calculate 25th, 50th, and 90th percentile of ratings
p25 = np.percentile(overall_array, 25)
p50 = np.percentile(overall_array, 50)
p90 = np.percentile(overall_array, 90)

print("25th Percentile:", p25)
print("50th Percentile (Median):", p50)
print("90th Percentile:", p90)
```

```
25th Percentile: 62.0
50th Percentile (Median): 66.0
90th Percentile: 75.0
```

In [60]:

```python
# Business-Oriented Insights

print("BUSINESS INSIGHTS & RECOMMENDATIONS")

#  Top countries
top_country = fifa['nationality'].value_counts().idxmax()
print(f"Most players come from {top_country}, indicating a mature football system. "
      "Clubs and scouts should focus here for large talent pools.")

# Average ratings by country
avg_rating_country = fifa.groupby('nationality')['overall'].mean().sort_values(ascending=False).head(5)
best_country = avg_rating_country.index[0]
print(f" Players from {best_country} have the highest average ratings, suggesting scouting here may yield top-tier talent.")

#  Age performance peak
peak_age = fifa.groupby('age')['overall'].mean().idxmax()
print(f" Player performance peaks around age {peak_age}. Clubs should prioritize players in their mid-20s to early 30s for max

#  Wage insights
highest_paid = fifa[['short_name', 'wage_eur']].sort_values('wage_eur', ascending=False).iloc[0]
print(f" {highest_paid['short_name']} is the highest paid player (€{highest_paid['wage_eur']}/week), "
      "reflecting the commercial and performance value of elite players.")

#  Attribute correlation
cols = ['overall', 'shooting', 'passing', 'dribbling', 'defending', 'physic']
existing_cols = [c for c in cols if c in fifa.columns]  # only use columns that exist

if len(existing_cols) > 1:
    correlation = fifa[existing_cols].corr()['overall'].drop('overall').sort_values(ascending=False)
    top_skill = correlation.index[0]
    print(f" {top_skill.capitalize()} has the strongest correlation with overall rating, "
          "so training and scouting should prioritize this skill.")
else:
    print(" Correlation could not be calculated because 'overall' or other skill columns are missing.")

#  Club insights
```

```
top_club = fifa.groupby('club')['overall'].mean().sort_values(ascending=False).idxmax()
print(f" {top_club} has the highest average player rating, showing strong recruitment strategies worth studying.")
```

```
BUSINESS INSIGHTS & RECOMMENDATIONS

Most players come from England, indicating a mature football system. Clubs and scouts should focus here for large talent pools.
 Players from Mozambique have the highest average ratings, suggesting scouting here may yield top-tier talent.
 Player performance peaks around age 41. Clubs should prioritize players in their mid-20s to early 30s for maximum impact.
 L. Messi is the highest paid player (€565000/week), reflecting the commercial and performance value of elite players.
 Passing has the strongest correlation with overall rating, so training and scouting should prioritize this skill.
 FC Bayern München has the highest average player rating, showing strong recruitment strategies worth studying.
```

In [ ]:

In [ ]: